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JOURNAL OF NEUROTRAUMA 36:650–660 (March 1, 2019)
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DOI: 10.1089/neu.2018.5739

Longitudinal Resting State Functional Connectivity Predicts Clinical Outcome in Mild Traumatic Brain Injury

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Abstract

Mild traumatic brain injury (mTBI) affects about 42 million people worldwide. It is often associated with headache, cognitive deficits, and balance difficulties but rarely shows any abnormalities on conventional computed tomography (CT) or magnetic resonance imaging (MRI). Although in most mTBI patients the symptoms resolve within 3 months, 10–15% of patients continue to exhibit symptoms beyond a year. Also, it is known that there exists a vulnerable period post-injury, when a second injury may exacerbate clinical prognosis. Identifying this vulnerable period may be critical for patient outcome, but very little is known about the neural underpinnings of mTBI and its recovery. In this work, we used advanced functional neuroimaging to study longitudinal changes in functional organization of the brain during the 3-month recovery period post-mTBI. Fractional amplitude of low frequency fluctuations (fALFF) measured from resting state functional MRI (rs-fMRI) was found to be associated with symptom severity score (SSS, $r = -0.28$, $p = 0.002$). Decreased fALFF was observed in specific functional networks for patients with higher SSS, and fALFF returned to higher values when the patient recovered (lower SSS). In addition, functional connectivity of the same networks was found to be associated with concurrent SSS, and connectivity immediately after injury (<10 days) was capable of predicting SSS at a later time-point (3 weeks to 3 months, $p < 0.05$). Specific networks including motor, default-mode, and visual networks were found to be associated with SSS ($p < 0.001$), and connectivity between these networks predicted 3-month clinical outcome (motor and visual: $p < 0.001$, default-mode: $p < 0.006$). Our results suggest that functional connectivity in these networks comprise potential biomarkers for predicting mTBI recovery profiles and clinical outcome.

Keywords: connectivity; fractional amplitude of low frequency fluctuations (fALFF); longitudinal; predict; resting state fMRI

Introduction

Concussion, generally classified as mild traumatic brain injury (mTBI), is typically associated with axonal damage resulting from a head injury.[1] Although its true incidence is unknown, mTBI related to sports injuries alone can be estimated to occur in more than half a million Americans annually.[2–4] mTBI has become an increasingly recognized public health concern facing significant scrutiny.[5] Following mTBI, patients often exhibit quantifiable neurological symptoms that typically include headache, fatigue, depression, anxiety, irritability, and impaired cognitive and balance function.[6] These symptoms are non-specific and their neuroanatomical localization is often unclear.

Whereas in most mTBI patients, symptoms resolve within 3 months post-injury, 10–15% of the patients continue to exhibit symptoms beyond 1 year.[7,8] Further, cumulative brain injury is a rising concern among mTBI patients; during recovery, a second injury may lead to chronic cognitive impairment,[9] and hence it is critical to identify this vulnerable window for appropriate treatment planning. However, objective measures of diagnosis, severity, and prognosis remain poorly validated in mTBI.[10]

Conventional computed tomography (CT) and anatomical magnetic resonance imaging (MRI) techniques are generally insensitive to subtle changes in mTBI, and do not provide the information to prognose mTBI recovery profile.[11–13] Microhemorrhages in TBI are often conspicuous in susceptibility-weighted images (SWI).[14–16]

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There is evidence that TBI pathoanatomical features observed in acute TBI (using T1 IR-spoiled gradient recalled [SPGR], T2, T2 fluid-attenuated inversion recovery [FLAIR], T2*-weighted gradient recalled echo [GRE]) improves predictive power of 3-month outcomes, compared with clinical and demographic/socioeconomic factors alone, or in combination with CT imaging.[17] However, mTBI pathology is often too subtle to be detected by conventional radiographic or MRI techniques in the majority of the disease population.

Advanced neuroimaging may offer improved, objective characterization of the injury for prognosis and treatment planning.[18,19] Some reports of mTBI implicate structural damage, which manifested in behavioral symptoms.[6] Structural white matter (WM) changes measured using quantitative diffusion have been found to correlate with cognitive performance, with marked reduction in fractional anisotropy (FA) thought to reflect axonal injury.[20–22] But there is limited reproducibility and poor technical standardization across studies with respect to affected WM regions.[22,23] This may be because structural damage is heterogeneous due to the mechanism of the injury, even though common behavioral symptoms are observed.[6] Given the consistency in behavioral symptoms, assessment of brain function may prove to be an important measure for evaluating brain–behavior relationship in mTBI. Resting state functional MRI (rs-fMRI) has been widely used to study brain–behavior relationships in various diseases including Alzheimer's disease and schizophrenia.[24–27] Prior work in mTBI using rs-fMRI has shown abnormal functional connectivity in the default-mode network (DMN)[28–30] as well as other resting state networks including fronto-parietal and motor networks.[31–34]

Whereas previous studies have largely been cross-sectional in design, there are a few reports of longitudinal studies with moderate sample size that have shown changes in perfusion,[35] structural,[30] and functional connectivity[28,30,31] in mTBI patients as they recover over months to 1-year post-injury. There is a need for large, systematic, prospective longitudinal studies, with data sampled from acute time-points (within hours of injury) to months post-injury to better understand the dynamic nature of mTBI recovery. Such longitudinal data would provide information about changes in the brain with disease progression and recovery, and help identify biomarkers that could be used for prognosis. In this study, we followed mTBI patients from acute time-points (<10 days) to 3 months post-injury and acquired functional neuroimaging and behavioral symptomatology at each clinical encounter to (1) identify functional neuroimaging features related to post-concussive symptoms, and to (2) use features from acute encounters post-injury to predict patient outcome at late encounters (>15 days).

Methods

Subject sample

A total of 91 mTBI patients were included in the study after obtaining informed consent from three sites (Hospital for Special Surgery, New York; Houston Methodist Hospital, Houston, Texas; and University of California at San Francisco). Each patient underwent neurological and neuropsychological testing in addition to MRI scans post-injury for up to four clinical encounters (<3 days [E1], 5–10 days [E2], 15–30 days [E3], 83–103 days [E4]; Fig. 1A). For the data analysis, early encounters were termed as encounters E1–E2 and late encounters were termed as encounters E3–E4. Twenty-three healthy controls were also recruited, and the same measurements were obtained at two encounters 10–21 days apart (Table 1). The mTBI patients and healthy control participants were age and gender matched (Table 1). For this study, we recruited patients with closed-head injuries including sports accidents (non-professional and professional), accidental falls, automobile accidents, and bicycle accidents. Enrolled patients were between the ages of 15 and 51 years at the time of enrollment and were diagnosed with mTBI (Table 1).

Exclusion criteria included any of the following: (1) loss of consciousness >15 min after injury, (2) post-traumatic amnesia lasting >24 h after TBI event, (3) scoring <13 on the Glasgow Coma Scale (GCS), (4) structural brain injury indicated by previous neurological findings, (5) previous history of moderate or severe TBI, (6) a previous mTBI within 12 months, (7) previously diagnosed brain WM disease, (8) history of seizures within past 10 years, (9) history of self-reported recreational drug usage in past 10 years, (10) history of alcohol abuse or dependence (per the *Diagnostic and Statistical Manual of Mental Disorders* [DSM]-IV-TR Diagnostic Criteria), (11) current primary Axis I or II psychiatric disorder, except for disorders classified as minor and not expected to impact study conduct or integrity, (12) history of brain mass, neurosurgery, stroke, or dementia, (13) known cognitive dysfunction, structural brain disease, or malformation, or (14) current anti-psychotic, psychotropic, or anti-epileptic medication usage. Formal assessment was conducted based on DSM-IV-TR Diagnostic Criteria. If the participant was suspected to have alcohol abuse or dependency, their use was assessed by the clinician along with the coordinator and compared with the DSM definitions.

Imaging protocol and behavioral assessment

Identical MRI scanners and coils were used at each location with identical scanning protocols, and were tested for uniformity. Using 3T MRI scanners (GE Signa MR750; GE Healthcare, Chicago, IL) and 32-channel head coil (Nova Medical, Wilmington, MA), rs-fMRI was acquired using multi-band (acceleration factor = 3) two-dimensional (2D) echo-planar imaging (EPI), repetition time (TR)/echo time (TE) = 900/30 msec, flip angle = 70 degrees for 6 min (395 volumes), with 1.875 mm^2 in-plane resolution and 3 mm slice thickness (no slice gap using interleaved slice order) to cover the whole brain. Three-dimensional (3D) T1-weighted SPGR BRAVO scan (1 mm isotropic resolution) was acquired at each encounter, which was used for normalization purposes. Additional image sequences were acquired, but their discussion is outside the scope of the current study. Physiological measures of pulse rate and blood pressure were also taken at each session. At each session, the patient completed the Sport Concussion Assessment Tool-2 (SCAT-2), a questionnaire administered by physicians and clinical staff.[36]

Statistical analysis

Resting state fMRI analysis. For each subject, rs-fMRI data were first pre-processed, and then fractional amplitude of low frequency fluctuations (fALFF) and seed-based connectivity (SBC) were calculated.

Pre-processing. Resting state fMRI data were motion corrected, rigidly registered (6 degrees of freedom) to the T1-weighted image data, non-rigidly registered to Montreal Neurological Institute (MNI) atlas via the T1-weighted image data (Fig. 1B). For fALFF analysis, the motion-corrected and co-registered rs-fMRI data were spatially smoothed using a 3D Gaussian filter (FWHM 4 mm, Fig. 1B). For SBC analysis, the motion-corrected and co-registered rs-fMRI data underwent additional pre-processing including physiological nuisance removal using a CompCor[37] (motion parameters, mean WM, mean ventricular cerebrospinal fluid [CSF], and global signal were used), spatial smoothing using a Gaussian filter (FWHM 4 mm), and temporal band-pass filtering (0.01–0.1 Hz) (Fig. 1B). All the pre-processing was performed using custom-built software in C++ and Python.[38] We included

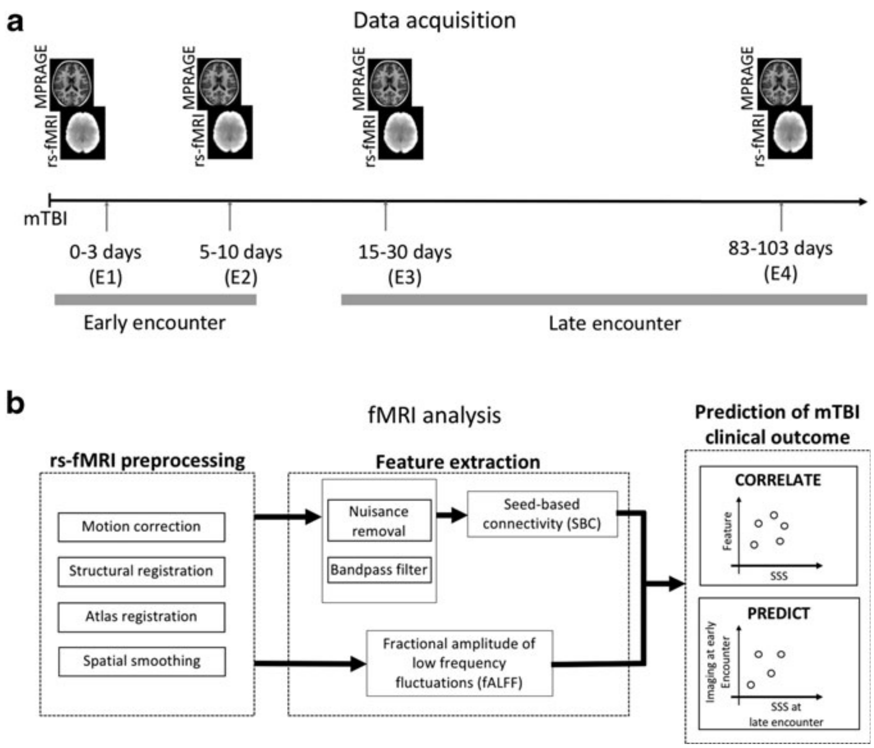

FIG. 1. Neuroimaging data acquisition and analysis. **(A)** Longitudinal imaging protocol. Neuroimaging data were collected at up to four clinical encounters from less than 3 days to 3 months post-mTBI (<3 days [E1], 5–10 days [E2], 15–30 days [E3], 83–103 days [E4]). **(B)** Conceptual diagram of resting state fMRI pre-processing and analytics. fMRI, functional magnetic resonance imaging; mTBI, mild traumatic brain injury.

global signal regression during pre-processing due to several advantages associated with global signal regression including better specificity of positive correlations and removal of nuisance signal associated with motion, cardiac, and respiratory signals.[39–42]

The rs-fMRI and T1-weighted image data were visually inspected for gross artifacts. Because excessive head motion is a concern for rs-fMRI, data from subjects with excessive head motion (motion >3 mm between two successive volumes) were excluded. Twelve encounters from mTBI patients and controls were not included in the analysis due to excessive motion. The numbers in Table 1 and analysis do not include these 12 encounters. For fALFF and SBC analysis a subset of the data that had all the necessary information was included, that is, subjects whose behavioral or physiological measures were missing were discarded from analysis that required these variables (Table 1).

Fractional amplitude of low frequency fluctuations (fALFF). fALFF was calculated from pre-processed rs-fMRI data as the ratio of power in low-frequency band (0.01–0.1 Hz) to power of the entire frequency range (0–0.55 Hz).[43] Correlation of each voxel's fALFF with concurrent Symptom Serverity Score (SSS) was computed for every encounter. Repeat measurements (multiple measurements from the same subject at different encounters) were adjusted using linear mixed models shown in Figure 2. Age, gender, years of education, blood pressure (systolic/diastolic), and pulse rate were used as co-variates of no-interest in the model. In addition, mean fALFF was calculated for each of the 14 functional networks derived from 90 functional regions of interest (ROI).[44] fALFF values of the 14 networks were correlated with concurrent SSS across all subjects. *P*-values cited in text and Figure 3 are after Bonferroni correction used to correct for multiple comparisons. The number of subjects used for correlation analysis

Table 1. Demographics of Included Participants

Subset	N	*Encounter*	*M/F*	*Age*	*Score*
All	114 (mTBI: 91; HC: 23)	218 (mTBI: E1 = 28; E2 = 67; E3 = 54; E4 = 41; Controls: E1 = 16; E2 = 12)	60/54	24.2 ± 8.7	20.1 ± 23.2
fALFF (correlation)	91	190 (E1 = 28; E2 = 67; E3 = 54; E4 = 41)	49/42	23.4 ± 8.8	34.7 ± 23
SBC (correlation)	90	90 (E1 = 27; E2 = 51; E3 = 9; E4 = 3)	49/41	23.4 ± 8.8	34.5 ± 23
SBC (prediction)	53	53	28/25	23.4 ± 8.6	21.7 ± 22.9[a]
Controls	23	28	11/12	25.1 ± 8.7	0.04

[a]Score at E3 post mTBI.

fALFF, fractional amplitude of low-frequency fluctuations; HC, healthy control; mTBI, mild traumatic brain injury; SBC, seed-based connectivity.

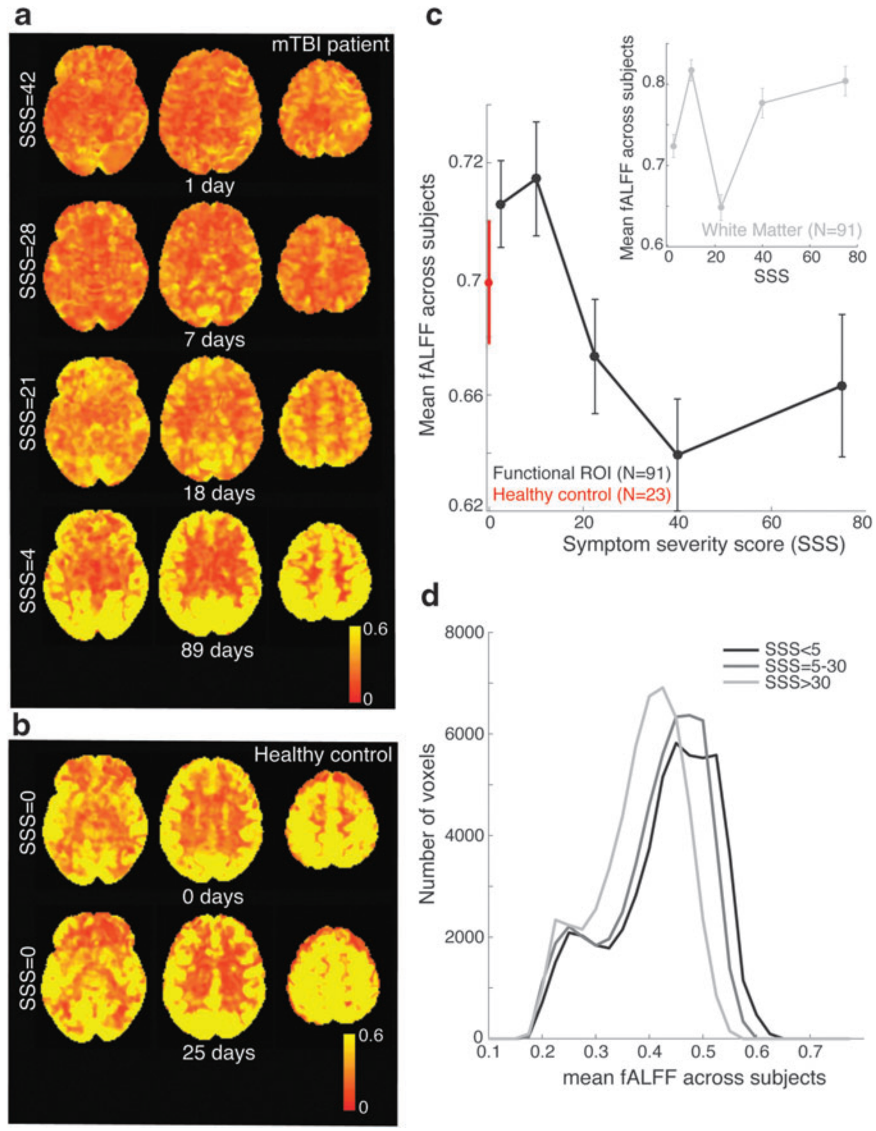

FIG. 2. fALFF in whole-brain gray matter (GM) was significantly correlated with SSS in mTBI. **(A)** fALFF map of example mTBI patient (patient clinically recovered over 3 months, 16-year-old male). fALFF in GM increased with patient recovery (lower SSS). Color bar indicates fALFF. Brighter colors indicate higher fALFF. **(B)** fALFF map of example healthy control (HC) subject at two separate encounters (37-year-old female). fALFF in HC was high at both encounters and comparable to values at E4 (89 days) for mTBI patient shown in (A). **(C)** fALFF was significantly correlated with SSS ($r=-0.28$, $p=0.002$, $n=91$ patients, linear mixed model). This correlation was not observed in white matter (WM) ($p=0.18$, inset). **(D)** fALFF distribution was significantly different for low (SSS <5) and high (SSS >30) SSS patient groups ($p<0.001$, ks-test). fALFF, fractional amplitude of low-frequency fluctuations; mTBI, mild traumatic brain injury; SSS, symptom severity score.

was 91 (Table 1). All the fALFF analysis was performed using custom-built code in MATLAB (Mathworks, Natick, MA).

Seed-based connectivity (SBC). Twelve seed-based functional connectivity maps[45] were computed from 6-mm spherical ROIs (Table 2). The seeds were obtained from previously published locations[24,46,47] or were hand-drawn on the atlas using well-established anatomical landmarks. For each seed, mean time courses were computed from the pre-processed rs-fMRI data. Pearson correlation of the mean time courses of each seed with every voxel in the brain was computed to produce the corresponding functional connectivity map. These correlation maps were then transformed to z-scores using Fisher's z-transform to produce 12 functional network maps for each subject.

For each network, correlation of functional connectivity z-score with concurrent SSS was computed for only one encounter for each subject (Table 1). It was corrected for multiple comparisons using $p=0.001$ with cluster level thresholding, using SPM12 (http://www.fil.ion.ucl.ac.uk/), with permutation testing to give an overall p-value <0.05 (Figs. 4, 5, Tables 3, 4). This was performed with only one data point per subject to avoid non-unique measurements (multiple data from the same subject). Whereas multiple comparison across voxels was corrected for each network, given the limited number of patients, correction for multiple networks was not performed for SBC analysis. For prediction analysis, SBC at the early encounters (earlier of E1/E2) was used to predict SSS at the late encounters post-injury (earlier of E3/E4) using Pearson correlation. This was corrected for multiple comparisons. Age, gender,

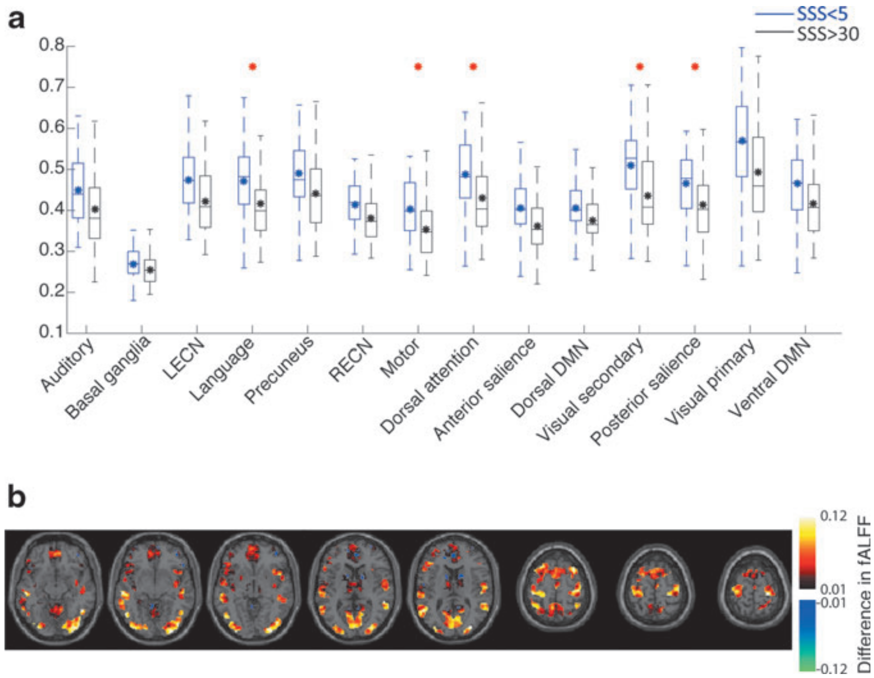

FIG. 3. Changes in fALFF with symptom severity scores (SSS) across functional networks. **(A)** Motor, salience, language, and secondary visual networks showed significant differences in fALFF between patients with low (SSS <5, blue line) and high (SSS >30, black line) SSS (*$p<0.03$, Bonferroni corrected. The p-value without correction was $p<0.03/14$). **(B)** Difference in fALFF between low (SSS <5) and high (SSS >30) groups across 90 ROIs. Larger differences (brighter colors) were observed in the visual and motor regions. DMN, default-mode network; fALFF, fractional amplitude of low-frequency fluctuations; LECN, left executive control network; RECN, right executive control network; ROIs, regions of interest.

years of education, blood pressure (systolic/diastolic), and pulse rate were used as co-variates of no-interest in the general linear model (GLM) for correlation and prediction analysis. The number of subjects used for this predictive analysis was 53 (Table 1, only included subjects with imaging data at early encounter and SSS at the late encounter). In addition, prediction analysis was repeated for (1) encounter E4 only ($n=36$ mTBI patients with imaging data at early encounter and E4 SSS), and (2) after accounting for the effects of early SSS in the prediction of late SSS to account for correlation between early and late encounter SSS. Early SSS was added as a ‘‘covariate of no interest'' in the GLM.

Results

We collected rs-fMRI data from mTBI patients ($n=91$, age $=23.4\pm8.8$, 42 females) (Fig. 1A, Table 1) at four encounters, E1–E4 (Supplementary Fig. 1; see online supplementary material at http://www.liebertpub.com).

Neuropyschological measures

SSS were computed from the SCAT-2 questionnaire and ranged from 0 to 132. Detailed demographic and related SSS information

TABLE 2. SEED LOCATIONS OF 12 FUNCTIONAL NETWORKS USED IN SEED-BASED CONNECTIVITY ANALYSIS

		MNI coordinates (mm)		
Network	*Seed region*	*X*	*Y*	*Z*
DMN	PCC	2	–54	26
DAS	IPS R	24	–60	50
ECN (L)	Left DLPFC	–42	34	20
ECN (R)	Right DLPFC	44	36	20
Motor dorsal (L)	Left precentral gyrus hand knob	–28	–26	64
Motor dorsal (R)	Right precentral gyrus hand knob	34	–24	60
Motor ventral (L)	Left precentral gyrus ventral	–56	–6	24
Motor ventral (R)	Right precentral gyrus ventral	60	–2	24
Salience (L)	Left anterior insula	–32	26	–14
Salience (R)	Right anterior insula	38	22	–10
Visual primary	BA17	8	–78	8
Visual secondary	BA18	–22	–90	2

Networks: DAS, dorsal attention network; DMN, default-mode network; ECN, executive control networks.
Seeds: BA, Broadmann area; DLPFC, dorso-lateral prefrontal cortex; IPS, intraparietal sulcus; PCC, posterior cingulate cortex.

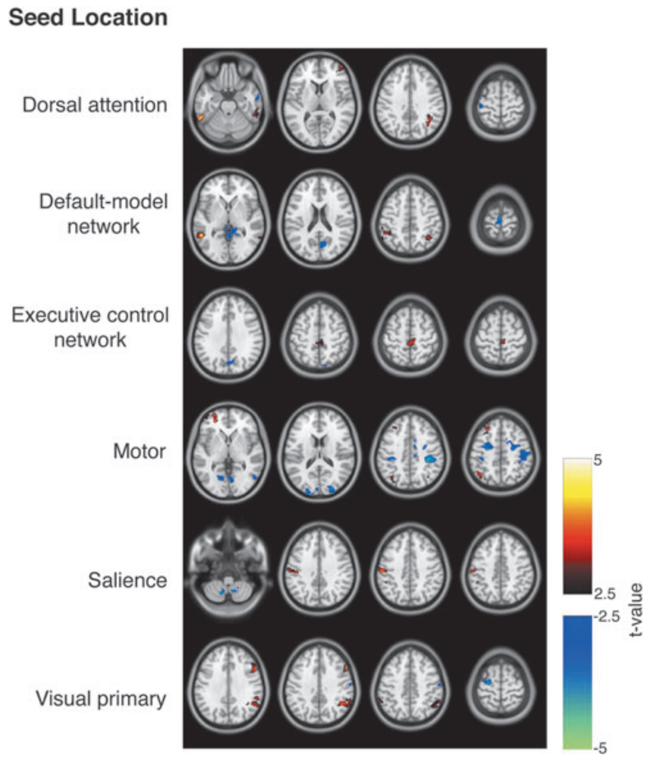

FIG. 4. SBC connectivity was significantly correlated with concurrent SSS. Statistically significant results of seeds of dorsal attention network, default mode network, executive control network, motor, salience, and visual network are shown ($p < 0.05$, cluster threshold = 50 voxels). Warm colors (red-yellow) are regions with connectivity that positively correlate with SSS. Cold (blue-cyan) are regions with connectivity that negatively correlate with SSS. SBC, seed-based connectivity; SSS, symptom severity score.

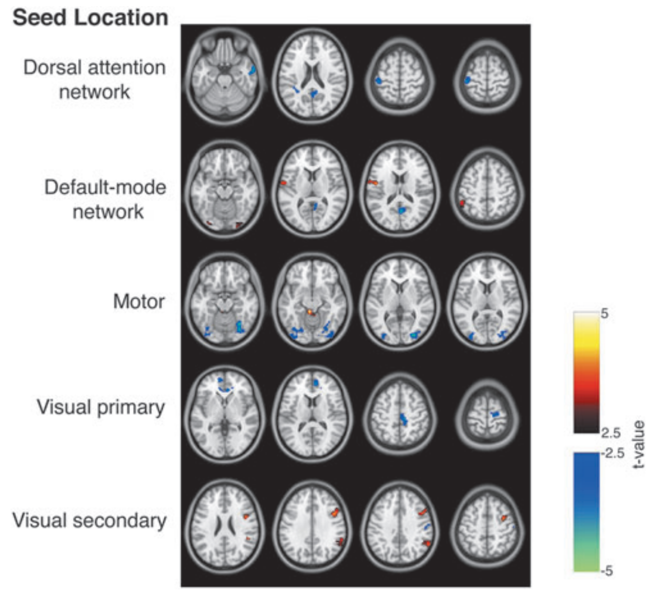

FIG. 5. SBC at early encounters predicted SSS at late encounters. Connectivity in default mode network, dorsal attention network, motor network, and visual network at early encounter significantly correlates with (predicts) score at late encounter ($p < 0.05$, cluster threshold = 50 voxels). Statistically significant positive and negative correlations are shown in warm colors (red-yellow) and cold colors (blue-cyan) respectively. SBC, seed-based connectivity; SSS, symptom severity score.

is provided in Table 1. SSS for 22/23 healthy controls was 0, with one subject with SSS = 1. Across patients, SSS decreased over the course of 3 months post-injury (Supplementary Fig. 2; see online supplementary material at http://www.liebertpub.com), and 73% of patients had SSS <10 by 3 months. The average SSS at E1 (<3 days post-injury) was 34.4 ± 20.2 ($n = 28$), E2 (5–10 days post-injury) was 33.7 ± 25.1 ($n = 67$), E3 (15–30 days) was 16.1 ± 20.01 ($n = 54$), and E4 (83–103 days post-injury) was 9.3 ± 16.2 ($n = 41$). The most common symptoms observed at the early encounters (E1/E2) included headache, not feeling right, feeling slowed down, and fatigue. Demographic information and physiological variables including age, years of education, pulse rate, blood pressure, and motion parameters were not significantly correlated to SSS (Supplementary Figs. 3, 4; see online supplementary material at http://www.liebertpub.com).

Imaging measures

Eighty-nine of the 91 (98%) patients included in the study demonstrated no anatomical changes attributable to mTBI on conventional anatomical MR images, which included high-resolution 3D susceptibility and diffusion-weighted imaging. Two patients showed anatomical changes at the early encounters, as identified by a neuroradiologist: one patient showed a few small clustered subcortical WM hemorrhages within the right centrum

Table 3. Correlation between Functional Connectivity Derived Using Seed-Based Connectivity and Symptom Severity Score

Seed	*Region of interest (ROI)*	*Correlation with concurrent symptom severity score*
DAS	Fusiform gyrus (left + right)	+
	Motor	–
	DAS	+
DMN	IPL, DLPFC	+
	DAS	+
	Visual primary	–
ECN	Motor	+
Motor	Visual primary	–
	Motor	–
	DLPFC	+
	ECN (left)	+
Salience	Motor	+
	Cerebellum	–
Visual	Motor	–
	ECN (right)	+

(+) indicates significant positive correlation ($p < 0.05$, family-wise error [FWE] corrected) and (–) indicates significant negative correlation ($p < 0.05$, FWE corrected).

Networks: DAS, dorsal attention network; DMN, default-mode network; ECN, executive control networks.

Seeds: DLPFC, dorso-lateral prefrontal cortex; IPS, intraparietal sulcus; PCC, posterior cingulate cortex.

Table 4. Correlation between Functional Connectivity Derived Using Seed-Based Connectivity at Early Encounters and symptom severity score (SSS) at Late Encounters

Seed	*ROI*	*Prediction of E3 SSS*
DMN	Motor	–
	Visual primary	+
	DAS	+
DAS	Motor	–
Motor	Visual secondary	–
Salience	Anterior cingulate	–
Visual	Motor	–
	Anterior cingulate	–
	ECN (right)	+

(+) indicates significant positive correlation ($p < 0.05$, family-wise error [FWE] corrected) and (–) indicates significant negative correlation ($p < 0.05$, FWE corrected).

Networks: DMN, default-mode network, DAS, dorsal attention network, ECN, executive control networks.

semiovale, and another showed a small focal hemorrahagic lesion with surrounding edema within the left body of the corpus callosum. These findings were both interpreted as acute hemorrhagic axonal injuries.

Whole brain analysis of fALFF as a function of symptom severity

fALFF was significantly associated with SSS (Fig. 2). Figure 2A shows an example mTBI patient with trend of fALFF increase with patient clinical recovery (lowering of SSS) over 3 months. Patient showed SSS = 42 (1 day post-injury) and low fALFF values across brain regions at E1 compared with SSS = 4 and higher fALFF at E4 (89 days) with clinical recovery. The fALFF values at E4 were similar to healthy control (Fig. 2B). Healthy controls showed higher values of fALFF in gray matter (GM) regions compared with WM regions as seen in Figure 2B. To assess repeatability in healthy controls, test-retest analysis was conducted for healthy controls showing repeatability (intra-class coefficient [ICC] = 0.68, Supplementary Methods; see online supplementary material at http://www.liebertpub.com) across scans (Supplementary Fig. 5; see online supplementary material at http://www.liebertpub.com).

At a group level, mean fALFF in GM regions was significantly negatively correlated with SSS ($r = -0.28$, $p = 0.002$; Fig. 2C). In the group analysis, because each mTBI patient was recorded at multiple encounters, a linear mixed effects model was used to account for repeated measures. Adding to the specificity of this correlation, fALFF in WM regions was not found to be correlated with SSS (inset, Fig. 2C). mTBI patients were categorized into three groups: low SSS (SSS <5), intermediate (SSS >5 and <30) and high SSS (SSS >30). The distribution of mean fALFF values in GM across subjects was significantly shifted to lower values for high SSS ($p < 0.001$, ks-test, Fig. 2D). fALFF was not significantly different between healthy controls and low SSS mTBI patients ($p > 0.05$, rank sum test). Patients with high SSS had significantly lower fALFF than controls (fALFF in high SSS patients = 0.39 ± 0.07, fALFF in controls = 0.42 ± 0.06, $p = 0.02$, rank sum test). Although the SSS decreased over the course of 3 months post-injury, fALFF values were not significantly correlated with time after injury ($p > 0.05$).

Network-level analysis of fALFF as a function of symptom severity

We compared fALFF in 14 functional networks between groups with low and high SSS (Fig. 3A). Motor, visual, language, and salience networks showed significantly higher fALFF at low compared with high SSS (Fig. 3B, $p < 0.03$, rank sum test, Bonferroni corrected). On comparing the changes in fALFF between patient groups with low (SSS <5) and high (SSS >30) scores, different degrees of change were observed across the 90 functional ROIs, with the largest difference between the two groups observed in regions within motor and visual networks (Fig. 3B). In contrast to the ROI-based analysis, voxel-based analysis at the whole-brain level did not show any clusters with significant correlation to SSS after correcting for multiple comparisons.

To assess whether early time-point fALFF could predict outcome in mTBI, we calculated correlation between fALFF at early encounters (E1–E2) with SSS at late encounters (E3–E4). Early encounter fALFF was not significantly correlated to late encounter SSS, at voxel or network-level.

SBC analysis as a function of symptom severity

Resting state functional networks were extracted using SBC (see section, ''Methods'') and healthy control data showed typical resting state networks (Supplementary Fig. 6; see online supplementary material at http://www.liebertpub.com). Functional connectivity was correlated with concurrent SSS in several regions of specific networks, including the dorsal attention, default mode, executive control (ECN), motor, visual, and salience networks (Fig. 4, Table 3). For example, dorsal attention network seed connectivity was positively correlated with SSS in areas including fusiform face area, and negatively correlated with SSS in primary motor regions. Motor, visual, and the DMNs were found to be associated strongly with SSS ($p < 0.05$, cluster threshold). We did not detect significant connectivity associated with SSS in networks other than those mentioned in Table 3. Additionally, independent component analysis (ICA) was used to further validate the SBC maps' correlation with SSS. Between network connectivities measured as correlations between ICA-derived network time courses showed similar network connectivity associations as observed using SBC (Supplementary Table 1; see online supplementary material at http://www.liebertpub.com). However, within network functional connectivity derived using ICA component maps did not yield any significant correlation with SSS (Supplementary Fig. 7; see online supplementary material at http://www.liebertpub.com).

In addition to correlating connectivity to concurrent SSS at all encounters, we estimated association between connectivity at early encounters (E1–E2) and SSS at late encounters (E3–E4) (see section, ''Methods'' and Fig. 1). Regions with statistically significant correlation (prediction) of late encounter SSS (SSS at E3 or E4) with early encounter (E1 or E2) functional connectivity for several networks are shown in Figure 5 and listed in Table 4. For example, DMN seed connectivity was positively correlated with SSS in visual and inferior parietal lobule (IPL) regions and negatively correlated in primary motor regions. Here again, visual, motor, and DMN networks seem to be strongly predicting late encounter SSS. The same networks were associated with predicting late encounter SSS, on repeating the analysis with SSS at E4 only (Supplementary Table 2; see online supplementary material at http://www.liebertpub.com). To account for the correlation between early and late encounter SSS (Supplementary Fig. 2), prediction analysis was repeated after accounting for early encounter SSS as a covariate of no-interest in the GLM. Identical networks were involved after accounting early encounter SSS (Supplementary Table 3; see online supplementary material at http://www.liebertpub.com). Within-network connectivity derived using ICA maps at early encounters did not yield any significant correlation with symptom severity scores at the late encounters (Supplementary Fig. 7). However, ICA-derived between network connectivities prediction of late encounter SSS showed similar results as SBC (Supplementary Tables 1, 4; see online supplementary material at http://www.liebertpub.com).

Discussion

We examined longitudinal functional rs-fMRI biomarkers that are associated with mTBI symptoms with the eventual goal of predicting patient outcome and recovery. Previous fMRI research,including rs-fMRI, has examined the neurophysiological underpinnings of mTBI and its utility for potential treatment approaches.[28,32–34,48] We followed patient recovery by measuring longitudinal changes in rs-fMRI activity from less than 3 days and up to 3 months post-injury. We used multi-band rs-fMRI acquisition and a 32-channel head coil to provide high spatio-temporal resolution data. This acquisition protocol has been shown to give high sensitivity and improved signal-to-noise ratio (SNR) over traditional fMRI sequences,[49,50] allowing us to investigate subtle functional changes associated with mTBI. We used cutting-edge multi-band protocol with sub-second ($TR = 900$ msec) whole brain acquisition providing more than twice as many data points as conventional fMRI (typical $TR = 2$ sec) providing improved sensitivity and allowing for shorter scan times.To our knowledge, this is the first study of this magnitude to track mTBI patients over time using advanced acquisition fMRI to study changes associated with clinical recovery. We observed that fALFF in rs-fMRI was negatively correlated with SSS (Figs. 2, 3). This change was most prominent in the motor and visual regions. Further, we observed significant correlation between functional connectivity in the motor, visual, and DMNs and patient recovery profiles (Figs. 4, 5), providing evidence that these spatially resolved functional differences may be useful as an index of patient recovery and treatment monitoring, as well as toward understanding the neural underpinnings of mTBI.

Multiple comparison corrections in fMRI has come under critical scrutiny in the recent past due to potential lenient methods used to account for the large number of comparisons made across voxels.[51] In this study, to overcome such errors, we have used up-to-date cluster size threshold implemented in SPM12. In addition, we also compared the cluster size threshold in SPM with the much stricter permutation testing method to verify the results obtained.

Previous rs-fMRI studies have discovered both reduction and increase in connectivities of functional brain networks after mTBI.[18,28] In agreement with our study, decreases in DMN functional connectivity have been found in mTBI and post-concussive patients.[28,32,33] The majority of these studies have used ICA-derived functional connectivity to assess group differences between acute mTBI patients and healthy controls. Here, we characterized longitudinal changes in functional connectivity as a function of recovery profiles based on SSS from 3 days to 3 months post-injury. In concordance with previous studies, significant differences between healthy controls and early encounter mTBI patients were observed in fALFF and functional connectivity (Fig. 2A). Further, we observed low values of fALFF in mTBI patients with high symptom severity, gradually increasing to values close to healthy controls as patients recovered (Fig. 2). This change in fALFF was more prominent in specific networks including motor, visual, and salience that correspond to symptoms of visuo-motor dysfunction (balance, blurriness) and cognitive deficits (attention, memory) typically observed in early encounters of mTBI.[52,53] Future work includes probing visual, motor, and higher cognition-related sub-symptom scores association with the corresponding networks. Functional connectivity between the same networks (visual, motor, salience, and ECN) correlate with SSS. Previously, we have shown that the connectivity between these networks decreases in mTBI patients with high symptom severity returning to higher values (close to healthy controls) during patient recovery[38] showing that both fALFF and functional connectivity follow recovery profiles in mTBI. Palacios and colleagues showed decreased functional connectivity in visual and ventral DMN regions in patients who presented more post-concussive symptoms.[31] We show that mTBI symptoms are strongly correlated with between-network connectivity especially between the motor, visual, and DMN networks. Further, we asked if we can predict late encounter SSS from neuroimaging at an early encounter. We find that functional connectivity between motor, visual, and DMN at early encounters predict symptom severity at late encounters (Fig. 5, Table 4), showing evidence that rs-fMRI can be used as a tool to predict long-term outcome in mTBI.

fALFF in visual, motor, and salience networks was negatively correlated with concurrent SSS. This is in agreement with previous studies that have shown significantly reduced amplitude of low-frequency fluctuations in regions including frontal, temporal, and occipital cortices in acute mTBI patients.[34,54] Because fALFF is weakly dependent on SNR, it is possible that changes in fALFF could arise from changes in non-neuronal activity, such as motion and other physiological noise. However, we observed no correlation between motion parameters and SSS (Supplementary Fig. 4). In addition, the observed correlation between fALFF and SSS cannot be attributed purely to non-neuronal sources because (1) differential changes were observed across functional ROIs (Fig. 3), indicating that the observed changes were not from a global noise source, (2) the rate of change of recovery was different in each network/functional ROI, and (3) physiological measurements (Supplementary Fig. 3) and fALFF in WM regions was not correlated with SSS (Fig. 2C). Whereas voxel-level analysis of fALFF did not show significant correlation to SSS after appropriate multiple comparison corrections, significant correlation was observed between fALFF and SSS at a functional ROI/network level. This discrepancy could be because of loss of power due to multiple comparisons at the voxel level and that averaging over functionally homogeneous regions increased power in network-level analysis. In support of the above, we have previously shown that compared with functional ROIs described in this study, similar analysis using structurally derived ROIs did not show significant correlation between fALFF and SSS.[55]

To further interrogate functional changes at the network level, we calculated SBC for 12 typical functional networks and correlated changes in connectivity to SSS. In concordance with fALFF, connectivity in the same networks observed in fALFF analysis including salience, motor, and visual networks,were correlated to SSS (Fig. 4). In addition, functional connectivity between these networks predicted SSS at late encounters (E3/E4). Prediction analysis using only the 3-month outcome (E4) also showed similar results (Supplementary Table 2) showing *long-term* prognostic value of functional connectivity. In addition to SBC (which uses pre-determined seed locations to derive functional networks), functional connectivity was also estimated using ICA, a blind-decomposition method to extract spatially independent functional regions.[56] Individual ICA networks did not show any correlation with SSS (Supplementary Fig. 7). SBC has been show to measure total connectivity (between and within network connectivity), whereas ICA separates between-network connectivity from within network connectivity.[57] The lack of correlation between individual ICA network maps and SSS most likely indicates that mTBI severity is reflected in connectivity between networks, and not within networks, at least for the milder spectrum of TBI pathology. For example, connectivity within the motor network (or visual network) was not significantly correlated with SSS, but connectivity between motor and visual networks was negatively correlated with SSS. We have shown previously that the changes in SSS were more strongly reflected in between-network compared than in within-network connectivity features[38] indicating that mTBI disrupts communication between brain network modules.

Both SBC and fALFF showed correlation with SSS in the motor, visual, and salience networks. Although it is tempting to presume that fALFF and SBC provide the same information in the context of mTBI, this is not the case because (1) previously, we have shown that fALFF could explain only the positive correlation in SBC networks,[58] and (2) fALFF is a within-voxel measure, whereas SBC measures relationship between brain voxels. Although negative correlations in functional connectivity are not fully understood, this has been shown to be related to brain function[24,25] and disease, including schizophrenia,[59] attention deficit hyperactivity disorder (ADHD),[60] and Alzheimer disease.[61]

Although our study demonstrates novel rs-fMRI biomarkers for mTBI clinical outcome prediction, the study does have limitations. The difference in encounter times between healthy controls and mTBI patients could lead to differences in the two groups. To match the sampling times, fALFF analysis was repeated without including data from 3-month encounter post-injury. Even after removing the 3-month encounters, fALFF was significantly correlated to SSS ($r=-0.21$, $p=0.013$). SCAT-2 is a questionnaire that might be affected by patient reporting. Future studies will include objective measurements of symptoms in addition to SCAT-2. Also, we combined all behavioral symptoms into one common score (SSS), whereas our results clearly show involvement of specific functional networks in our mTBI cohort (Figs. 3, 4). Further, each patient showed severity of a different symptom, and combining them into one score might dilute patient variability. Future studies will include specific sub-scores and cognitive tests related to visual, motor, and attention with the corresponding networks, to understand the neural mechanisms and contributions of these networks at an individual patient level. Different injury types can lead to different regional brain damage and symptomatology. Although we acknowledge this heterogeneity, for simplicity we grouped all the patients in our analysis. Hence our results perhaps focus only on common symptoms that are prevalent across various injury types. To address the contribution of heterogeneity to the observed results, two methods were used for subgrouping the mTBI patient population to reduce heterogeneity (see online Supplementary Methods): (1) patients were grouped into subgroups based on late encounter SSS; fALFF in the late encounter SSS >10 patient subgroup showed higher correlation to SSS compared with Figure 2C (Supplementary Fig. 8A,B; see online supplementary material at http://www.liebertpub.com); and (2) the subgroup of mTBI patients with fALFF much lower than healthy controls also showed stronger negative correlation with SSS ($r=-0.39$, see online Supplmentary Methods). The most prominent symptoms within this subgroup were headache, don't feel right, and fatigue (Supplementary Fig. 8C). In our patient cohort, only 2 of 91 patients showed high symptom severity (SSS >30) at late encounters. For clinical mTBI outcome prediction, inclusion of more patients with chronic symtamology will be important to build robust models. Further, given that mTBI is primarily a structural damage, future clinical outcome prediction models will involve combining data from advanced structural and functional neuroimaging.

Behavioral measures of mTBI severity are highly subjective and patient dependent. Symptomatology and current structural neuroimaging are not sufficiently sensitive to allow the detection of subtle brain changes that occur after mTBI. Here, we show that rs-fMRI can be used as an objective measure to assess symptom severity and can be used as a biomarker for predicting symptom severity after brain injury. Further validation with larger datasets, accounting for the disease heterogeneity, could allow for the use of rs-fMRI analysis as a prognostic tool for mTBI. We suggest that neuroimaging can be used in conjuction with clinical assessment to enable an objective measurement to the evaluation of mTBI patients, leading to patient stratification and identification of patients who may require long-term intervention.

Acknowledgments

We thank the General Electric Company (GE)/National Football League (NFL) Medical Advisory Board for their comments and advice. At GE, we thank Dr. Victor Miranda and Amy Gallenberg for their role in the execution of the study, and the neuro MRI team, in particular Pauline Worters, Matthew Middione, and Ajit Shankaranarayanan, for MRI pulse sequence design. We would also like to thank the technical and clinical staff at Hospital for Special Surgery, Houston Methodist Hospital, and University of California, San Francisco (specifically Nicholas Pojman, Nhu Nhu Nguyen, and Chelsea Camara) for their role in patient recruitment and data acquisition. The clinical staff at Hospital for Special Surgery included Kelianne Cummings, Lorena Loci, Dan Zuchelli, Megan Parmenter, and Kristin Halvorsen, all of whom were integral in recruiting patients and collecting data. Finally, we thank the GE/NFL Head Health Initiative (HHI) for sponsoring and funding the study.

Author Disclosure Statement

The study was sponsored by GE and funded jointly by GE and the NFL under the HHI. Radhika Madhavan, Suresh Joel, Rakesh Mullick, and Luca Marinelli are employees of GE. Teena Shetty, Pratik Mukherjee, and Joseph Masdeu have received grant support from GE and are members of the GE/NFL HHI medical advisory board. Teena Shetty has financial relationships with ChemBio/Perseus, Teva Pharmaceuticals, and ElMindA Ltd.

References

1. Langlois, J.A., Rutland-Brown, W., and Wald, M.M. (2006). The epidemiology and impact of traumatic brain injury: a brief overview. J. Head Trauma Rehabil. 21, 375–378.
2. O'Kane, J.W., Spieker, A., Levy, M.R., Neradilek, M., Polissar, N.L., and Schiff, M.A. (2014). Concussion among female middle-school soccer players. JAMA Pediatr. 168, 258–264.
3. Rosenthal, J.A., Foraker, R.E., Collins, C.L., and Comstock, R.D. (2014). National high school athlete concussion rates from 2005–2006 to 2011–2012. Am. J. Sports Med. 42, 1710–1715.
4. Theadom, A., Starkey, N.J., Dowell, T., Hume, P.A., Kahan, M., McPherson, K., Feigin, V., and BIONIC Research Group. (2014). Sports-related brain injury in the general population: an epidemiological study. J. Sci. Med. Sport 17, 591–596.

5. Grool, A.M., Aglipay, M., Momoli, F., Meehan, W.P., Freedman, S.B., Yeates, K.O., Gravel, J., Gagnon, I., Boutis, K., Meeuwisse, W., Barrowman, N., Ledoux, A.-A., Osmond, M.H., Zemek, R., and Pediatric Emergency Research Canada (PERC) Concussion Team. (2016). Association between early participation in physical activity following acute concussion and persistent postconcussive symptoms in children and adolescents. JAMA 316, 2504–2514.
6. McInnes, K., Friesen, C.L., MacKenzie, D.E., Westwood, D.A., and Boe, S.G. (2017). Mild traumatic brain injury (mTBI) and chronic cognitive impairment: a scoping review. PLoS One 12, e0174847.
7. Daneshvar, D.H., Riley, D.O., Nowinski, C.J., McKee, A.C., Stern, R.A., and Cantu, R.C. (2011). Long term consequences: effects on normal development profile after concussion. Phys. Med. Rehabil. Clin. N. Am. 22, 683–700.
8. Rabinowitz, A.R., and Levin, H.S. (2014). Cognitive sequelae of traumatic brain injury. Psychiatr. Clin. North Am. 37, 1–11.
9. Covassin, T., Moran, R., and Wilhelm, K. (2013). Concussion symptoms and neurocognitive performance of high school and college athletes who incur multiple concussions. Am. J. Sports Med. 41, 2885–2889.
10. Carney, N., Ghajar, J., Jagoda, A., Bedrick, S., Davis-O'Reilly, C., du Coudray, H., Hack, D., Helfand, N., Huddleston, A., Nettleton, T., and Riggio, S. (2014). Executive summary of Concussion guidelines step 1: systematic review of prevalent indicators. Neurosurgery 75, Suppl. 1, S1–S2.
11. Iverson, G.L. (2005). Outcome from mild traumatic brain injury. Curr. Opin. Psychiatry 18, 301–317.
12. Kurca, E., Sivák, S., and Kucera, P. (2006). Impaired cognitive functions in mild traumatic brain injury patients with normal and pathologic magnetic resonance imaging. Neuroradiology 48, 661–669.
13. Le, T.H., and Gean, A.D. (2009). Neuroimaging of traumatic brain injury. Mt. Sinai J. Med. N. Y. 76, 145–162.
14. Haacke, E.M., Xu, Y., Cheng, Y.-C.N., and Reichenbach, J.R. (2004). Susceptibility weighted imaging (SWI). Magn. Reson. Med. 52, 612–618.
15. Benson, R.R., Gattu, R., Sewick, B., Kou, Z., Zakariah, N., Cavanaugh, J.M., and Haacke, E.M. (2012). Detection of hemorrhagic and axonal pathology in mild traumatic brain injury using advanced MRI: implications for neurorehabilitation. NeuroRehabilitation 31, 261–279.
16. Liu, W., Soderlund, K., Senseney, J.S., Joy, D., Yeh, P.-H., Ollinger, J., Sham, E.B., Liu, T., Wang, Y., Oakes, T.R., and Riedy, G. (2015). Imaging cerebral microhemorrhages in military service members with chronic traumatic brain injury. Radiology 278, 536–545.
17. Yuh, E.L., Mukherjee, P., Lingsma, H.F., Yue, J.K., Ferguson, A.R., Gordon, W.A., Valadka, A.B., Schnyer, D.M., Okonkwo, D.O., Maas, A.I.R., Manley, G.T., and TRACK-TBI Investigators. (2013). Magnetic resonance imaging improves 3-month outcome prediction in mild traumatic brain injury. Ann. Neurol. 73, 224–235.
18. Eierud, C., Craddock, R.C., Fletcher, S., Aulakh, M., King-Casas, B., Kuehl, D., and LaConte, S.M. (2014). Neuroimaging after mild traumatic brain injury: review and meta-analysis. Neuroimage Clin. 4, 283–294.
19. Jaiswal, M.K. (2015). Toward a high-resolution neuroimaging biomarker for mild traumatic brain injury: from bench to bedside. Front. Neurol. 6, 148.
20. Niogi, S.N., Mukherjee, P., Ghajar, J., Johnson, C., Kolster, R.A., Sarkar, R., Lee, H., Meeker, M., Zimmerman, R.D., Manley, G.T., and McCandliss, B.D. (2008). Extent of microstructural white matter injury in postconcussive syndrome correlates with impaired cognitive reaction time: a 3T diffusion tensor imaging study of mild traumatic brain injury. AJNR Am. J. Neuroradiol. 29, 967–973.
21. Niogi, S.N., Mukherjee, P., Ghajar, J., Johnson, C.E., Kolster, R., Lee, H., Suh, M., Zimmerman, R.D., Manley, G.T., and McCandliss, B.D. (2008). Structural dissociation of attentional control and memory in adults with and without mild traumatic brain injury. Brain J. Neurol. 131, 3209–3221.
22. Shenton, M., Hamoda, H., Schneiderman, J., Bouix, S., Pasternak, O., Rathi, Y., Vu, M.A., Purohit, M., Helmer, K., Koerte, I., Lin, A., Weston, C.F., Kikinis, R., Kubicki, M., Stern, R., and Zafonte, R. (2012). A review of magnetic resonance imaging and diffusion tensor imaging findings in mild traumatic brain injury. Brain Imaging Behav. 6, 137–192.
23. Dodd, A.B., Epstein, K., Ling, J.M., and Mayer, A.R. (2014). Diffusion tensor imaging findings in semi-acute mild traumatic brain injury. J. Neurotrauma 31, 1235–1248.
24. Fox, M.D., Snyder, A.Z., Vincent, J.L., Corbetta, M., Van Essen, D.C., and Raichle, M.E. (2005). The human brain is intrinsically organized into dynamic, anticorrelated functional networks. Proc. Natl. Acad. Sci. U. S. A. 102, 9673–9678.
25. Greicius, M.D., Krasnow, B., Reiss, A.L., and Menon, V. (2003). Functional connectivity in the resting brain: a network analysis of the default mode hypothesis. Proc. Natl. Acad. Sci. U. S. A. 100, 253–258.
26. Rashid, B., Damaraju, E., Pearlson, G.D., and Calhoun, V.D. (2014). Dynamic connectivity states estimated from resting fMRI Identify differences among schizophrenia, bipolar disorder, and healthy control subjects. Front. Hum. Neurosci. 8, 897.
27. Zhou, J., Greicius, M.D., Gennatas, E.D., Growdon, M.E., Jang, J.Y., Rabinovici, G.D., Kramer, J.H., Weiner, M., Miller, B.L., and Seeley, W.W. (2010). Divergent network connectivity changes in behavioural variant frontotemporal dementia and Alzheimer's disease. Brain 133, 1352–1367.
28. Mayer, A.R., Bellgowan, P.S.F., and Hanlon, F.M. (2015). Functional magnetic resonance imaging of mild traumatic brain injury. Neurosci. Biobehav. Rev. 49, 8–18.
29. Zhou, Y., Milham, M.P., Lui, Y.W., Miles, L., Reaume, J., Sodickson, D.K., Grossman, R.I., and Ge, Y. (2012). Default-mode network disruption in mild traumatic brain injury. Radiology 265, 882–892.
30. Dall'Acqua, P., Johannes, S., Mica, L., Simmen, H.-P., Glaab, R., Fandino, J., Schwendinger, M., Meier, C., Ulbrich, E.J., Müller, A., Baetschmann, H., Jäncke, L., and Hänggi, J. (2017). Functional and Structural Network Recovery after Mild Traumatic Brain Injury: A 1-Year Longitudinal Study. Front. Hum. Neurosci. 11, 280.
31. Palacios, E.M., Yuh, E.L., Chang, Y.-S., Yue, J.K., Schnyer, D.M., Okonkwo, D.O., Valadka, A.B., Gordon, W.A., Maas, A.I.R., Vassar, M., Manley, G.T., and Mukherjee, P. (2017). Resting-state functional connectivity alterations associated with six-month outcomes in mild traumatic brain injury. J. Neurotrauma 34, 1546–1557.
32. Shumskaya, E., Andriessen, T.M.J.C., Norris, D.G., and Vos, P.E. (2012). Abnormal whole-brain functional networks in homogeneous acute mild traumatic brain injury. Neurology 79, 175–182.
33. Stevens, M.C., Lovejoy, D., Kim, J., Oakes, H., Kureshi, I., and Witt, S.T. (2012). Multiple resting state network functional connectivity abnormalities in mild traumatic brain injury. Brain Imaging Behav. 6, 293–318.
34. Zhou, Y., Lui, Y.W., Zuo, X.-N., Milham, M.P., Reaume, J., Grossman, R.I., and Ge, Y. (2014). Characterization of thalamocortical association using amplitude and connectivity of fMRI in mild traumatic brain injury. J. Magn. Reson. Imaging JMRI 39, 1558–1568.
35. Meier, T.B., Bellgowan, P.S.F., Singh, R., Kuplicki, R., Polanski, D.W., and Mayer, A.R. (2015). Recovery of cerebral blood flow following sports-related concussion. JAMA Neurol. 72, 530–538.
36. McCrory, P., Meeuwisse, W., Johnston, K., Dvorak, J., Aubry, M., Molloy, M., and Cantu, R. (2009). Consensus Statement on Concussion in Sport: The 3rd International Conference on Concussion in Sport held in Zurich, November 2008. J. Athl. Train. 44, 434–448.
37. Behzadi, Y., Restom, K., Liau, J., and Liu, T.T. (2007). A component based noise correction method (CompCor) for BOLD and perfusion based fMRI. Neuroimage 37, 90–101.
38. Ravishankar, H., Madhavan, R., Mullick, R., Shetty, T., Marinelli, L., and Joel, S.E. (2016). Recursive feature elimination for biomarker discovery in resting-state functional connectivity, in: *2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pps. 4071–4074.
39. Fox, M.D., Zhang, D., Snyder, A.Z., and Raichle, M.E. (2009). The global signal and observed anticorrelated resting state brain networks. J. Neurophysiol. 101, 3270–3283.
40. Weissenbacher, A., Kasess, C., Gerstl, F., Lanzenberger, R., Moser, E., and Windischberger, C. (2009). Correlations and anticorrelations in resting-state functional connectivity MRI: a quantitative comparison of preprocessing strategies. Neuroimage 47, 1408–1416.
41. Birn, R.M. (2012). The role of physiological noise in resting-state functional connectivity. Neuroimage 62, 864–870.
42. Power, J.D., Mitra, A., Laumann, T.O., Snyder, A.Z., Schlaggar, B.L., and Petersen, S.E. (2014). Methods to detect, characterize, and remove motion artifact in resting state fMRI. Neuroimage 84, 320–341.
43. Zou, Q.-H., Zhu, C.-Z., Yang, Y., Zuo, X.-N., Long, X.-Y., Cao, Q.-J., Wang, Y.-F., and Zang, Y.-F. (2008). An improved approach to detection of amplitude of low-frequency fluctuation (ALFF) for resting-state fMRI: fractional ALFF. J. Neurosci. Methods 172, 137–141.
44. Shirer, W.R., Ryali, S., Rykhlevskaia, E., Menon, V., and Greicius, M.D. (2012). Decoding subject-driven cognitive states with whole-brain connectivity patterns. Cereb. Cortex 22, 158–165.

45. Biswal, B., Yetkin, F.Z., Haughton, V.M., and Hyde, J.S. (1995). Functional connectivity in the motor cortex of resting human brain using echo-planar MRI. Magn. Reson. Med. 34, 537–541.
46. De Luca, M., Beckmann, C.F., De Stefano, N., Matthews, P.M., and Smith, S.M. (2006). fMRI resting state networks define distinct modes of long-distance interactions in the human brain. Neuroimage 29, 1359–1367.
47. Seeley, W.W., Menon, V., Schatzberg, A.F., Keller, J., Glover, G.H., Kenna, H., Reiss, A.L., and Greicius, M.D. (2007). Dissociable intrinsic connectivity networks for salience processing and executive control. J. Neurosci. 27, 2349–2356.
48. Palacios, E.M., Sala-Llonch, R., Junque, C., Roig, T., Tormos, J.M., Bargallo, N., and Vendrell, P. (2012). White matter integrity related to functional working memory networks in traumatic brain injury. Neurology 78, 852–860.
49. Preibisch, C., Castrillón G., J.G., Bührer, M., and Riedl, V. (2015). Evaluation of multiband EPI acquisitions for resting state fMRI. PLoS One 10, e0136961.
50. Todd, N., Moeller, S., Auerbach, E.J., Yacoub, E., Flandin, G., and Weiskopf, N. (2016). Evaluation of 2D multiband EPI imaging for high-resolution, whole-brain, task-based fMRI studies at 3T: sensitivity and slice leakage artifacts. Neuroimage 124, 32–42.
51. Eklund, A., Nichols, T.E., and Knutsson, H. (2016). Cluster failure: why fMRI inferences for spatial extent have inflated false-positive rates. Proc. Natl. Acad. Sci. U. S. A. 113, 7900–7905.
52. Katz, D.I., Cohen, S.I., and Alexander, M.P. (2015). Chapter 9: Mild traumatic brain injury, in: J. Grafman, and A.M. Salazar. (eds). *Handbook of Clinical Neurology*. Elsevier, pps. 131–156.
53. McCrea, M., Kelly, J.P., Randolph, C., Cisler, R., and Berger, L. (2002). Immediate neurocognitive effects of concussion. Neurosurgery 50, 1032–1040; discussion 1040–1042.
54. Zhan, J., Gao, L., Zhou, F., Bai, L., Kuang, H., He, L., Zeng, X., and Gong, H. (2016). Amplitude of low-frequency fluctuations in multiple-frequency bands in acute mild traumatic brain injury. Front. Hum. Neurosci. 10, 27.
55. Madhavan, R., Joel, S.E., Mullick, R., Shetty, T., and Marinelli, L. (2016). Functionally defined ROIs are superior to structurally defined ROIs for functional connectivity analysis in mild traumatic brain injury (mTBI). Abstract presented at the Fifth Biennial Conference on Resting State Brain Connectivity, Vienna, Austria.
56. Calhoun, V.D., Adali, T., Pearlson, G.D., and Pekar, J.J. (2001). A method for making group inferences from functional MRI data using independent component analysis. Hum. Brain Mapp. 14, 140–151.
57. Joel, S.E., Caffo, B.S., van Zijl, P.C., and Pekar, J.J. (2011). On the relationship between seed-based and ICA-based measures of functional connectivity. Magn. Reson. Med. 66, 644–657.
58. Chachra, P., Madhavan, R., and Joel, S.E. (2017). Low Frequency fluctuations are associated with positive but not negative functional connectivity. Annual Meeting of Organization of Human Brain Mapping, Vancouver, Canada.
59. Whitfield-Gabrieli, S., Thermenos, H.W., Milanovic, S., Tsuang, M.T., Faraone, S.V., McCarley, R.W., Shenton, M.E., Green, A.I., Nieto-Castanon, A., LaViolette, P., Wojcik, J., Gabrieli, J.D.E., and Seidman, L.J. (2009). Hyperactivity and hyperconnectivity of the default network in schizophrenia and in first-degree relatives of persons with schizophrenia. Proc. Natl. Acad. Sci. U. S. A. 106, 1279–1284.
60. Castellanos, F.X., Margulies, D.S., Kelly, A.M.C., Uddin, L.Q., Ghaffari, M., Kirsch, A., Shaw, D., Shehzad, Z., Di Martino, A., Biswal, B., Sonuga-Barke, E.J.S., Rotrosen, J., Adler, L.A., and Milham, M.P. (2008). Cingulate-precuneus interactions: a new locus of dysfunction in adult attention-deficit/hyperactivity disorder. Biol. Psychiatry 63, 332–337.
61. Wang, K., Liang, M., Wang, L., Tian, L., Zhang, X., Li, K., and Jiang, T. (2007). Altered functional connectivity in early Alzheimer's disease: a resting-state fMRI study. Hum. Brain Mapp. 28, 967–978.

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